TT-P-96D
June 11, 1976
SUPERSEDING
Fed. Spec. TT-P-96C
June 19, 1972

FEDERAL SPECIFICATION

PAINT, LATEX-BASE, FOR EXTERIOR SURFACES (WHITE AND TINTS)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers latex-base paint for application to exterior wood, masonry, metal and weathered paint substrates.

1.2 Classification.

1.2.1 Types. The latex base paint covered by this specification shall be furnished, ready-mixed, in the following types:

Type I	-	White.
II	-	Light tint.
III	-	Medium tint.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

P-R-550	-	Roller, Kit, Paint.
PPP-P-1892	-	Paint, Varnish, Lacquer, and Related Materials: Packaging, Packing and Marking of.
TT-P-25	-	Primer, Coating, Exterior (Undercoat for Wood, Ready-Mixed, White and Tints).

Federal Standards:

Fed. Test Method Std. No. 141	-	Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing.
Fed. Std. No. 595	-	Colors.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

FSC 8010

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:15Z
Check the source to verify that this is the current version before use.

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

D 358 - Specification for Wood to be Used as Panels in Weathering Tests for Paints and Varnishes.
D 476 - Test for Consistency of Paints Using the Stormer Viscosimeter.
D 562 - Test for COnsistency of Paints Using the Stormer Viscosimeter.
D 659 - Method of Evaluating Degree of Resistance of Chalking of Exterior Paints.
D 1210 - Test for Fineness of Dispersion of Pigment-Vehicle Systems.
D 1296 - Test for Residual Odor of Lacquer Solvents and Diluents.
D 1475 - Test for Density of Paint, Varnish, Lacquer and Related Products.
D 2369 - Test for Volatile Content of Paints.
D 3335 - Determination of Low Concentration of Lead in Paint by Atomic Absorption Spectroscopy.
E 97 - Test for 45-Degree Directional Reflectance of Opaque Specimens by Filter Photometry.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Materials. The manufacturer is given some latitude in the selection of materials and process of manufacture , provided that the paint supplied meets the requirements of this specification. Material shall be of the best quality used in good commercial practice and entirely suitable for the purpose intended under normal conditions of use. Material shall be free from all defects and imperfections that might affect serviceability of the finished product.

3.2 Composition.

3.2.1 Pigment. Any combination of prime pigments, including extenders, shall be good commercial quality, provided the paint meets all the requirements specified herein. For white, the opaque portion shall consist of titanium dioxide, rutile type, conforming to ASTM D 476, type IV. Not over six percent of anatase conforming to ASTM D 476, type I is permissible, and zinc oxide is permitted at discretion of the supplier provided the paint meets all the requirements. Such suitable extenders as mica (wet ground), calcium carbonate, silicates, and barytes may be used, provided all requirements are met. Tinting pigments may be used when necessary to match the color desired, provided these additional pigments have good color permanence. Toxic pigments are not allowed. A certified statement to this effect is necessary.

3.2.2 Vehicle. The vehicle shall consist of a water dispersion such as acrylate, vinyl chloride/acrylate, vinyl acetate/acrylate, and other plasticizing monomers such as maleates, fumarates, etc., together with such necessary additives as emulsifiers, pigment dispersants, anti-foaming agents, thickening or coalescing agents, and preservatives, provided the paint meets all the requirements of this specification.

3.2.3 Color. The color specified in the contract or purchase order (see 6.2) shall match the color chip in Fed. Std. No. 595, when tested as described in 4.4.3.

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:15Z
Check the source to verify that this is the current version before use.

3.3 Quantitative requirements. The quantitative requirements shall be as specified in tables I and II.

TABLE I. Quantitative requirements

Characteristics	Type I Min.	Type I Max.	Type II Min.	Type II Max.	Type III Min.	Type III Max.
Total solids, percent by weight of paint	57	--	55.5	--	54.5	--
Pigment, percent by weight of paint	--	38.5	--	37	--	35.5
Non-volatile vehicle, percent by weight of paint	20	--	19	--	19	--
Weight of paint (pounds per gallon)	11.7	--	11.7	--	11.4	--
Consistency (Krebs-Units)	75	97	75	95	75	95
Fineness of grind (Hegman scale)	3	--	3	--	3	--
Drying time:						
Set to touch, minutes	15	--	15	--	15	--
Dry hard, hour	--	1	--	1	--	1
Directional reflectance	86	--	--	--	--	--
Gloss 60° specular	8	15	8	15	8	15
Dry opacity (Contrast ratio)	0.92	--	--	--	--	--
Lead, percent non-volatile *	--	0.5	--	0.5	--	0.5

3.3.1 Dry opacity. Not more than one gallon per 630 square feet of the white paint, type I, (minimum apparent reflectivity 86 percent) shall be required to give a dry-film contrast ratio of 0.92. The minimum dry-film contrast ratio for tints applied to the same wet-film thickness in terms of apparent reflectivity shall be as shown in table II when tested as described in table IV.

TABLE II. Minimum dry film contrast ratios for tints

Apparent reflectivity of tints	Contrast ratio
Percent	
81 and above	0.92
77-80	0.93
73-76	0.94
69-72	0.95
65-68	0.96
61-64	0.97

3.4 Qualitative requirements.

3.4.1 Condition in container. The paint as received shall show no evidence of livering, skinning, or hard settling of the pigment. The container shall not be affected. The material shall be easily dispersed in the liquid portion by hand stirring to form a smooth mixture, free from persistant foam, when tested as specified in 4.4.4.

3.4.2 Storage stability. After 30 days storage in a half-filled 8-ounce screw-cap glass container, the paint shall be entirely suitable for use (no caking or hard settling) when tested as specified in 4.4.5. When stored in a closed container (quart plus or minus paint can) for 12 months the paint shall be useable with consistency change of not over five units from the original consistency.

3.4.3 Application. The paint, when tested as specified in 4.4.6, shall apply easily by brush, spray, or roller. There shall be no streaks, flashes, or laps. Brush marks shall not be conspicuous. The paint shall show good flowing and spreading properties. Leveling shall be characteristic for this type of paint.

3.4.4 Odor. During and after application, the odor of the paint shall be normal for the vehicle system used, when tested as specified in table IV.

3.4.5 Appearance. The paint, tested as specified in 4.4.7, shall dry to a uniform, smooth appearance when examined visually. The dried paint shall show no conspicuous brush marks or streaks.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:15Z
Check the source to verify that this is the current version before use.

3.4.6 Flexibility. The paint shall show no evidence of cracking, chipping, or flaking when subjected to the flexibility test specified in 4.4.8.

3.4.7 Freeze-thaw resistance. When tested as specified in 4.4.9, the paint shall show no coagulation or flocculation; shall be capable of being stirred into a smooth paint of uniform consistency; and the viscosity shall not be greater than 100 KU nor less than 70 KU.

3.4.8 Resistance to biological growth. When tested as specified in 4.4.10, the paint film shall show no biological growth.

3.4.9 Accelerated weathering. When tested as specified in 4.4.11, the paint film shall show no more chalking than that illustrated by pictorial standard No. 8, ASTM D 659. There shall be no other film defect detected by regular viewing of the panel.

3.4.10 Heat stability. When tested as described in 4.4.12, the paint shall not show significant seeding, coagulation, flocculation, or discoloration, and shall not exceed a viscosity of 100 KU.

3.4.11 Recoating properties. Upon recoating the paint surface after 6 hours of air drying of the first coat under standard laboratory conditions, no film irregularities shall be observed. There shall be no evidence of lifting or rolling of the previous coat when tested as described in 4.4.13.

3.4.12 Water resistance. When the paint film is tested as specified 4.4.14, the film shall show no blistering, wrinkling, re-emulsification, or other irregularities.

3.4.13 Washability (wet abrasion resistance). When painted panels are tested as in 4.4.15, the soiling mixture (synthetic) shall be substantially removed without any exposure of the undercoat.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Lot. For the purposes of sampling, a lot of the paint shall consist of a manufacturer's batch. A batch is defined as the end product of all raw materials mixed, blended, or processed in a single operation.

4.2.2 Sampling for inspection of filled containers. A random sample of filled containers shall be selected in accordance with MIL-STD-105 at inspection level I and acceptable quality level (AQL) = 2.5 percent defective to verify compliance with this specification regarding fill, closure, and marking and other requirements not involving tests.

4.3 Inspection.

4.3.1 Inspection of containers. Each sample filled container shall be examined for defects or construction of the container and the closure, for evidence of leakage, and for unsatisfactory markings; each filled container shall also be gaged to determine the amount to contents. Any container in the sample having one or more defects or under required fill shall be rejected, and if the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan of MIL-STD-105, the lot represented by the sample shall be rejected.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:15Z
Check the source to verify that this is the current version before use.

4.3.2 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the packaging, packing, and marking comply with the requirements of section 5 of this specification. Defects shall be scored in accordance with table III. For examination of interior packaging, the sample unit shall be one shipping container fully prepared for delivery selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure lsited shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 4.0 defects per hundred units.

TABLE III. Classification of preparation for delivery defects

Examine	Defects
Markings (exterior and interior)	Omitted; improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Bulging or distortion of container.

4.3.3 Testing of the end item. The methods of testing specified in 4.4 shall be followed. For purposes of sampling, the lot shall be expressed in units of gallons of paint. The sample unit for testing shall be one gallon of paint, randomly selected from containers in the lot. The paint shall be placed in separate clean, dry metal or glass containers, sealed, marked, and forwarded to the testing laboratories. The sample size shall be as follows:

Lot size (gallon)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and more	5

Tests shall be performed on a sample unit basis. All test reports shall contain the individual values utilized in expressing the final result. The lot shall be unacceptable if one or more sample unit fails to meet any test requirement specified.

4.4 Test Procedures.

4.4.1 The paint shall be tested in accordance with the applicable test methods indicated in table IV, and as specified hereinafter. Failure of any sample to pass any test and noncompliance with the requirements of this specification shall be cause for rejection of the lot represented by the sample. Unless otherwise specified, all tests shall be conducted at a temperature of 23 ± 1°C (73 ± 2°F) and a relative humidity of 50 ± 5%.

TABLE IV. Index.

Characteristics	Requirements Reference	Applicable Test Methods: Fed. Test Method Std. No. 141	Applicable Test Methods: ASTM Method
Total solids (nonvolatile content)	Table I	--	D 2369
Pigment content	Table I	4021	--
Nonvolatile vehicle	Table I	4053	--
Consistency	Table I	--	D 562
Drying time	Table I	4061.1	--
Fineness of grind	Table I	--	D 1210
Reflectance	Table I	--	E 97
Weight per gallon	Table I	--	D 1475
Dry opacity contrast ratio (procedure B)	Table I, II	4121	--
Odor	3.4.4	--	D 1296
Lead content	Table I	--	D 3335

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:15Z
Check the source to verify that this is the current version before use.

4.4.2 Pigment content. Weigh accurately to 0.1 mg an 8 to 10 g sample into a tared 50 ml centrifuge tube. Extract 3 times with 25 ml of a 1:9 acetone-water mixture. Centrifuge for 10 minutes at a relative Centrifugal Force (RCF) of approximately 3800. Decant the liquid portion carefully after each 10 minute extraction period. (The liquid extracted on the first three extractions will be slightly milky due to emulsifiers present in the paint.) The fourth extraction is made with 25 ml of acetone, followed by spinning for a 10-minute period at RCF of 2800. Decant the clear liquid portion. The fifth and sixth extractions are made using 25 ml of a 1:1 toluene: acetone mixture. Extraction time and speed are the same as that of the above. After the final extraction, the clear liquid portion is decanted and the tube is dried at 105 ± 3°C (221 ± 5°F) for 1 hour. The tube is cooled in a dessicator and weighed. If final pigment is hard and caked (indicating incomplete separation), repeat entire procedure using a new sample and an additional acetone-water extraction. Determine compliance with the requirement in Table I.

4.4.3 Color. Apply a coat of the paint on a clean, smooth, opaque, plate-glass panel using a suitable applicator that will deposit approximately a 25-um (0.001 inch) thick dry film. The paint film shall be allowed to dry for 24 hours at room temperature. Compare the color of the dried film in accordance with method 4250 of Fed. Test Method Std. No. 141, then observe for compliance with 3.2.3.

4.4.4 Condition in container. Examine the condition of the paint and the container as received in accordance with method 3011 of Fed. Test Method Std. No. 141, and evaluate for compliance with 3.4.1.

4.4.5 Storage stability. Half fill two 1-pint paint cans with thoroughly mixed paint. Close tightly and store in a dry place for 30 days. (Do not disturb the paint cans during the storage period.) After 30 days, examine the paint for compliance with 3.4.2. Fill a 1-quart paint can with thoroughly mixed paint and store for 12 months in a dry place. After 12 months, examine the paint for compliance with 3.4.2.

4.4.6 Application properties. The tests shall be conducted under standard laboratory conditions as specified in 4.4.1

4.4.6.1 Preparation of panels. The test panels shall be as follows:

1) Wood panel - Western-red-cedar, 13 x 46 x 2 cm (5 x 8 x 3/4 inch), prepared in accordance with ASTM method D 358.
2) Metal panel - Cold rolled steel, 13 x 46 x 0.09 cm (5 x 8 x 0.0375 inch), prepared in accordance with method 2011 of Fed. Test Method Std. No. 141.
3) Concrete - 15 x 15 x 5 cm (6 x 6 x 2 inch), prepared in accordance with Method 2051 of Fed. Test Method Std. No. 141.

A suitable primer shall be applied to the panels before application of the test paint.

4.4.6.2 Brushing. Place primed wood and primed steel panel in a vertical position and apply the paint at a spreading rate of approximately 350 to 450 square feet per gallon, using a 3-inch nylon brush which has been rinsed with water and shaken vigorously to remove excess water. With a well-filled brush, apply the paint by brushing across the panels, using back and forth strokes, and leveling up the coat with light strokes, using the tip of the brush. During the application note for pull, etc. Evaluate for compliance with 3.4.3.

4.4.6.3 Roller coating. Using a 7-inch roller complying to Fed. Spec. H-R-550, Class II, apply a coat of the paint to a primed wood and primed steel panel at the spreading rate of 450 square feet per gallon. Dry for 6 hours, and evaluate the finish for compliance with 3.4.3.

4.4.6.4 Spraying. Spray the paint on primed wood and primed steel panels using conventional or laboratory spray apparatus. Dry for 6 hours and examine for laps, flashes, etc. Evaluate for compliance with 3.4.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:15Z
Check the source to verify that this is the current version before use.

4.4.7 Appearance. Examine the dried film as applied in 4.4.6 by visual inspection, and evaluate for compliance with 3.4.5.

4.4.8 Flexibility. Apply a film of the paint to a flat tin panel (method 2012 of Fed. Test Method Std. No. 141) with a Bird applicator or other doctor blade to a wet-film thickness of 0.05 mm (0.002 inch). Air dry for 18 hours and bake for 3 hours at 105 ± 2°C (221 ± 4°F). Cool for 1/2 hour and bend over a 3.2 mm (1/8-inch) mandrel. Examine the film for cracks, chips, and flakes, and evaluate for compliance with 3.4.6.

4.4.9 Freeze-thaw resistance. Fill a one-pint friction-top can with the paint, as received, and close the can tightly. Expose the can and contents four times to the following temperature cycle:

(a) Low temperature of minus 18 ± 2°C (15 ± 3°F) for 16 hours.
(b) High temperatures of 25 ± 3°C (77 ± 5°F) for 8 hours.

After each exposure of the paint to the temperature cycle, examine it for coagulation or flocculation and determine the viscosity in accordance with ASTM method D 562. Evaluate for compliance with 3.4.7.

4.4.10 Resistance to biological growth. Determine the resistance to biological growth of the paint film in accordance with method 6271 of Fed. Test Method Std. No. 141. Aspergillus oryzae #10196 shall be used in the test. Test specimens shall not be leached before inoculation, and the inoculation period shall be seven days. All fungal mycelium and spores, slime and other biological growth shall be considered in the evaluation for compliance with 3.4.8.

4.4.11 Accelerated weathering. As the test surface, use a primed, flat tin panel prepared in accordance with method 2012 of Fed. Test Method Std. No. 141. Brush the paint at a spreading rate of 450 square feet per gallon on both sides and edges of the panel. Apply 2 coats, allowing 24 hours drying between coats and 72 hours before testing. Then expose the panel (primed side facing the arc) in accordance with method 6151 of Fed. Test Method Std. No. 141 for 300 hours. Evaluate for compliance with 3.4.9.

4.4.12 Heat stability. Mix the paint by hand to a uniform consistency. Fill a one-pint friction-top can with the paint and close the can tightly. Expose the can and contents to four cycles of alternate storage of 168 hours at 49 ± 2°C (120 ± 4°F) and then for 8 hours at room temperature. Evaluate for compliance with 3.4.10.

4.4.13 Recoating. After 6 hours air drying (see 4.4.6.2), apply the second coat of paint at a spreading rate of 450 square feet per gallon over one-half of each panel (see 4.4.6) in the same manner as the first coat. During application, note the working properties of the paint and behavior of the first coat. After 24 hours air drying, apply the second coat to the remaining portion of the panels at a spreading rate of 450 square feet per gallon. Dry for 6 hours and evaluate the finish as specified in 3.4.11.

4.4.14 Water resistance. Use the panel in 4.4.13 (recoating test) for this test. Place five drops of distilled water on the paint film and immediately cover the area with 50-mm watch glass. After 4 hours, remove the watch glass and note the appearance of the film. Remove the remaining water and allow 2 hours for recovery. Gently rub the film surface, and evaluate for compliance with 3.4.12.

4.4.15 Washability (wet abrasion resistance). Prepare the panels with an undercoat in accordance with method 6142 of Fed. Test Method Std. No. 141. Soil and test the panels according to method 6141 of Fed. Test Method Std. No. 141, except recharge the sponge every 25 cycles until a total of 750 cycles has been completed. Evaluate for compliance with 3.4.13.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:15Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. The paint shall be packaged, packed, and marked in accordance with PPP-P-.892. The level of packaging shall be level A, B, or C, and the level of packing shall be level A, B, or C as specified (see 6.2). The paint shall be furnished in 1-gallon metal cans or 5-gallon steel pails (see 6.2). All containers shall be resin-lined.

5.1.1 Additional marking. Each container shall also bear the following additional identification markings:

"Paint, Latex-Base for Exterior Surfaces (White and Tints)

"Instructions for use.

"Wood Surfaces.

"1. New wood: It is extremely important to thoroughly clean the surface before painting. Remove dirt, dust, etc. Sap streaks should be removed with turpentine or mineral spirits, and sap streaks and knots should be spot primed with a suitable sealer. Following spot priming, a blister-resistant primer or oil-base primer such as TT-P-25 should be applied if desired, followed by two coats of latex paint covered by this specification.

"2 Previously painted wood: Loose paint, as evidenced by flaking, cracking, and blistering, shall be completely removed, after which a coat of blister-resistant primer or paint shall be applied. This shall be followed by two coats of latex paint covered by this specification.

"Metal surfaces.

"Remove loose scale and rust by scraping, wire brushing, or sandblasting. Remove dust thoroughly. Prime metal with a suitable primer, and allow to dry hard before applying the latex paint.

"Masonry surfaces.

"No unusual preparation is required on new work, except cinder block and rough concrete surfaces should be grouted. On previously painted surfaces, all loose, powdery or flaking material, heavy chalking, or flaking and dirt should be removed by wire brushing. Then apply an alkali-resistant sealer, allow the sealer to dry hard, and apply the latex paint."

6. NOTES

6.1 Intended use. This specification covers a ready-mixed latex-base paint for exterior use on primed wood, primed metal, previously painted surfaces, and properly prepared masonry surfaces.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2.1).
(c) Color required (see 3.2.4).
(d) Size of container (see 5.1).
(e) Level of packaging and packing required (see 5.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:15Z
Check the source to verify that this is the current version before use.

6.3 Paint instructions. (see paragraph 5.1.1).

6.4 Unit of purchase. The unit of purchase shall be 1 U.S. gallon (which is equal to 3.785 liters or 231 cubic inches). Paint temperature shall not be in excess of 20°C (68°F) at the time of volume measurement.

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA-FSS
COM-NBS
VA-DMS
HUD-HAA
HEW-NIH
D.C. GOVT-DCG
DOT-RDS

Preparing activity:

GSA-FSS

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein. Price 35 cents each.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:15Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:15Z
Check the source to verify that this is the current version before use.